

JUAN BECERRA

Software Engineer – Gameplay, Animation & Systems

Seattle, WA | 📞 (208) 550-9191 | ✉ becerrajuan007@gmail.com | 🌐 hoowan.dev
🌐 [linkedin.com/in/hoowan](https://www.linkedin.com/in/hoowan) | 🐙 github.com/hoowan-dev

Experience

Electronic Arts – Lifestyle

2025 – Present

Software Engineer – Tools & Tech-Art

Kirkland, WA

- Architected an extensible Maya/Python animation stylization tool adopted across multiple internal studios, reducing redundant manual keyframing workflows by over 70%.
- Engineered a high-throughput Unreal Engine C++ plugin exporting runtime character trajectory, pose, and velocity data to train and populate motion-matching query databases.
- Prototyped and integrated ML-assisted authoring pipelines into DCC tools (leveraging PyTorch and ONNX runtime), enabling automated asset tagging and procedural pose generation without disrupting artist control.

Electronic Arts – Maxis Studios

2021 – 2025

Software Engineer – Gameplay & Animation

Seattle, WA (Remote)

- Architected a multi-threaded asynchronous path-planning and navigation API in UE5/C++ for *Project Rene (The Sims)*, enabling deterministic crowd routing and real-time obstacle avoidance within strict frame budgets (<1.5ms).
- Implemented core locomotion and runtime animation systems in UE5 (motion-matching, distance matching, custom IK solvers), delivering fluid, responsive character controls for next-gen simulation gameplay.
- Partnered directly with technical animators and gameplay designers to author in-editor tuning tools and visual debuggers, accelerating iteration cycles and authoring loops for complex behavior trees.
- Profiled and optimized runtime memory and frame budgets across character evaluation graphs and query sweeps, eliminating hitches during dense crowd simulation scenarios.

Lionbridge

2020 – 2021

Senior Games Test Engineer

Boise, ID

- Led a team of QA testers performing combinatorial and regression testing for live-service AAA titles, authoring technical repro steps and engine-level diagnostics in JIRA to accelerate developer triage.

Projects

Hoowan Game Engine | C++, OpenGL, EnTT, CMake | [GitHub](#)

Nov. 2020

- Engineered a custom C++ game engine featuring an OpenGL rendering submission pipeline, input event polling, and 2D collision detection and correction systems.
- Integrated an EnTT-based entity-component system (ECS) and constructed a companion desktop editor application for rapid scene composition and entity management.

DoomSlop | *Three.js*, *Web Audio API*, *Vite*, *JavaScript* | [Live Demo](#) | [GitHub](#)

Sep. 2026

- Built a 3D browser first-person shooter featuring hitscan raycasting, synthesized Web Audio sound effects, custom canvas radar minimap, and automated CI/CD deployments via GitHub Pages.

Technical Skills

Languages: C++ (Modern, Multi-threading), C#, Python, Rust, GLSL/HLSL, Java, JavaScript, SQL

Game Engines & APIs: Unreal Engine 5 (Source, Motion Matching, GAS, Mass/Crowd), Unity, Godot, Vulkan, OpenGL

Gameplay & Systems: Locomotion & Character Physics, Async Navigation/Pathfinding, Animation Runtime (IK, Motion-Matching), State Machines

Tools & ML Pipelines: Maya API, PySide/Qt, DCC Pipeline Automation, Unreal Plugin Development, PyTorch/ONNX, In-Editor Tooling

Infrastructure & Tools: Perforce (P4), Git, Docker, JIRA, CMake, Visual Studio, Linux, AWS, CI/CD

Education & Continuing Studies

Boise State University

2017 – 2021

Bachelor of Science in Computer Science

Boise, ID

University of Washington (Continuing Education)

2026 – Present

Graduate Coursework in Machine Learning & Deep Learning (Non-Degree, In Progress)

Seattle, WA